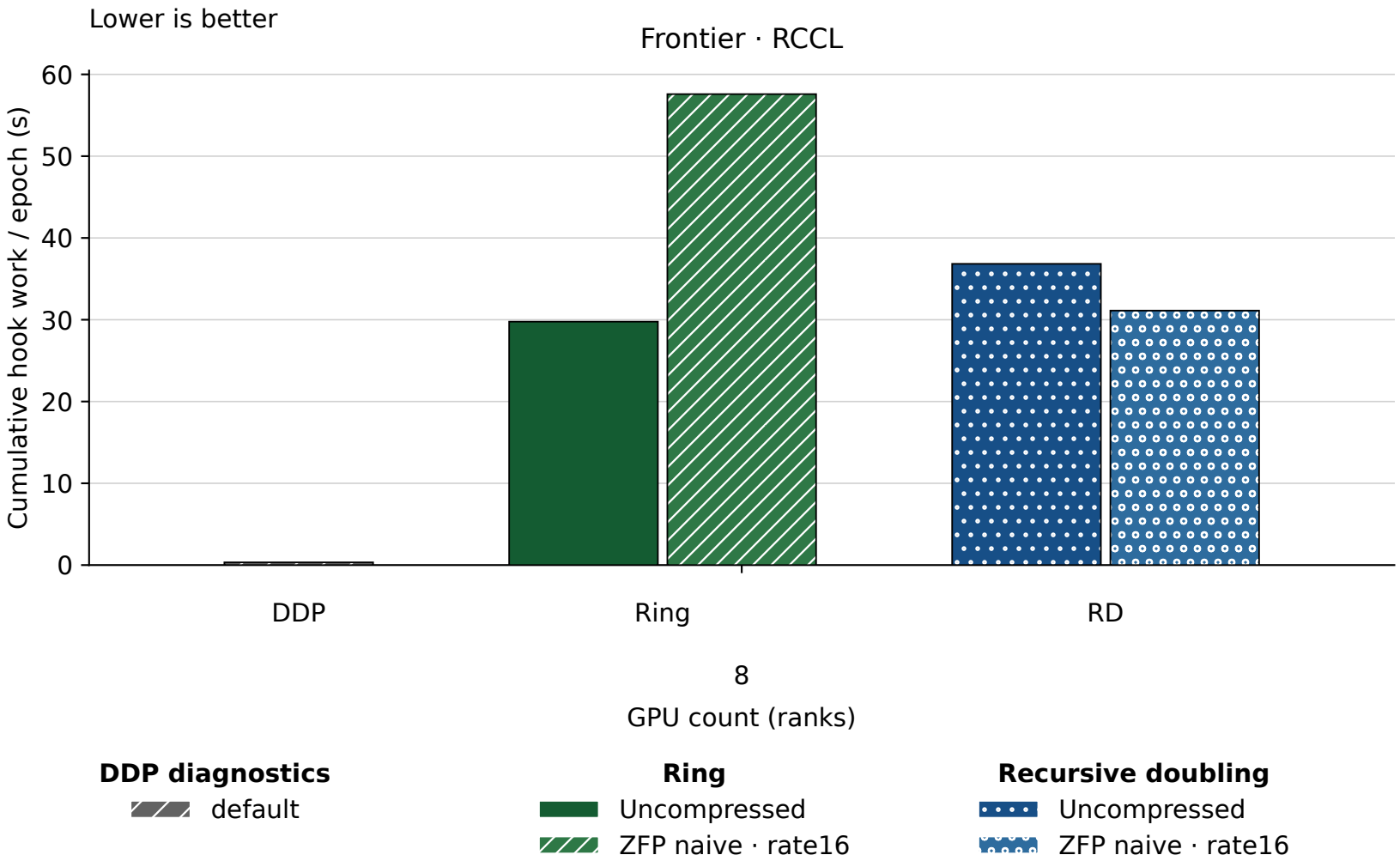

WideResNet 32×32 · strong scaling · effective global batch 128
Lower is better
Frontier · RCCL
Cumulative hook work / epoch (s)
60
50
40
30
20
10
0
DDP
Ring
RD
8
GPU count (ranks)
DDP diagnostics
default
Ring
Uncompressed
ZFP naive · rate16
Recursive doubling
Uncompressed
ZFP naive · rate16
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate